SaAr-DF at ArA-DF 2026: When More Data Hurts — Diversity-Driven Ensembling for Arabic Speech Deepfake Detection

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Abstract

We describe our system for Track 1 (dialect-generalization) of the ArA-DF-2026 Arabic speech deepfake detection shared task. Starting from the organizers' XLS-R+AASIST baseline (14.65% EER on the development test set), we show that (i) raw-waveform augmentation (RawBoost) roughly halves the error rate, (ii) replacing the AASIST back-end with a simple mean-pooling head over a post-trained self-supervised frontend removes a hard performance ceiling that persists even after full convergence, and (iii) a large, architecturally diverse ensemble is the only lever that keeps improving once single-model performance saturates. Our final system, a 13-member weighted ensemble spanning three frontend sizes, two back-ends, and four training-data recipes, reaches 2.34% EER on the development-test leaderboard and 2.42% EER on the official held-out blind evaluation set (144,210 utterances, 9× larger than the development test). We also report two negative results of practical value: injecting a large (54k-clip) multi-system Arabic text-to-speech corpus and a vocoder-copy-synthesis pseudo-spoof corpus both *degrade* accuracy, indicating that indiscriminately adding synthetic speech is harmful when its generation artifacts do not match the target domain. Our code is publicly available at https://github.com/Body123/aradf-2026-saar-df.

1 Introduction

Detecting synthetic ("spoofed") speech in Arabic is complicated by dialectal variation, code-switching, and a scarcity of large labeled deepfake corpora relative to English. The ArA-DF-2026 shared task (ArA-DF-2026 Organizing Committee, 2026) Track 1 targets *dialect generalization*: a Track-1 system trained on Egyptian, Gulf, and Modern Standard Arabic (MSA) speech is evaluated, in the blind phase, on a held-out test set almost an order of magnitude larger than the public development-test split, using Equal Error Rate (EER) as the sole, threshold-free metric.

We report a system-building process organized around three questions: (1) which architectural choices raise a *single* model's ceiling, (2) once that ceiling is reached, what further lowers error, and (3) which commonly-recommended interventions (extra synthetic data, signal-processing-based pseudo-spoof augmentation) actually help under this specific train/test distribution shift.

Our contributions are:

- An empirical ablation showing that back-end choice (AASIST vs. a plain mean-pooling head) dominates single-model performance once raw-waveform augmentation is applied, reversing the ranking observed before augmentation.
- Evidence that, once single-model performance saturates, weighted ensembling over architecturally *diverse* members is the only remaining lever with a reliable, repeated effect across our development cycle, even when individual added members are weaker in isolation.
- Two controlled negative results: a large multi-TTS-system external Arabic corpus and a vocoder-copy-synthesis pseudo-spoof corpus each *increase* error, both for a single model and inside the ensemble, despite targeting the task's known weak point (Gulf-dialect false accepts).

2 Related Work

Back-ends and frontends. AASIST (Jung et al., 2022) models spectro-temporal artifacts with a heterogeneous graph attention network and is a common back-end in anti-spoofing systems built on self-supervised speech encoders such as

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wav2vec 2.0 (Baevski et al., 2020) and its massively multilingual successor XLS-R (Babu et al., 2021). Truong et al. (2024) instead augment a Transformer/Conformer back-end with an explicit temporal-channel modeling (TCM) module. Ge et al. (2025) post-train wav2vec 2.0/HuBERT-family encoders specifically for deepfake detection on a very large multilingual corpus of genuine and artifact-bearing speech, and release these *post-trained* checkpoints (which we refer to as "AntiDeepfake" frontends).

Augmentation and data. RawBoost (Tak et al., 2022) augments raw waveforms with convolutive and additive noise to simulate channel and codec variation and is a standard component of ASVspoof-style pipelines. ASVspoof 2019 (Wang et al., 2020) and MLAAD (Müller et al., 2024) are widely used sources of non-Arabic and multilingual synthetic-speech attacks for co-training. For Arabic specifically, ArFake (Elsetohy et al., 2025) recently introduced a multi-dialect benchmark generated with four TTS/voice-conversion systems (XTTS-v2, FishSpeech, ArTST, VITS); we use a subset of this resource as an external data source (§4.3). Separately, vocoder-based copy-synthesis (resynthesizing genuine speech through a neural vocoder and relabeling it spoof) has been used to manufacture spoof training data cheaply (Wang and Yamagishi, 2023), but its benefit is contingent on the vocoder(s) used matching the target attacks' artifacts, and injecting such data at the supervised fine-tuning stage rather than as self-supervised pre-training can instead hurt performance (Wang and Yamagishi, 2024). More broadly, Müller et al. (2022) show that deepfake-detector generalization is highly sensitive to train/test artifact-distribution mismatch, the frame we adopt for our own negative results (§4.3).

Benchmarking and frameworks. Dowerah et al. (2025) benchmark open- and closed-source detectors across 14 evaluation sets and report that AASIST-based systems generalize comparatively poorly out-of-domain, motivating our back-end ablation in §4. We build our training pipeline on DeepFense (El Kheir et al., 2026), an open-source, modular framework that separates frontend, back-end, loss, and augmentation as interchangeable components.

3 System Description

3.1 Data

Our in-domain data is the ArA-DF-2026 Track-1 training split (Egyptian, Gulf, and MSA dialects). We co-train with three external attack sources: ASVspoof 2019 Logical Access (Wang et al., 2020) (non-Arabic synthetic/converted speech, for generic attack-type diversity), MLAAD's Arabic subset (Müller et al., 2024) (multilingual TTS attacks), and an Arabic-dialect spoof/bona-fide subset (dialect-diverse synthetic speech, including Gulf, our hardest dialect; §4). Every training recipe below is a named combination of these sources.

3.2 Frontend and back-end

We use post-trained self-supervised frontends from the AntiDeepfake family (Ge et al., 2025) at three sizes (300M/1B/2B parameters, built on MMS and XLS-R encoders), feeding either (a) AASIST (Jung et al., 2022), a graph-attention back-end used by the task baseline, or (b) a minimal mean-pooling ("Pool") head: temporal mean-pooling of the frontend's final hidden states followed by a single linear classification layer. Pool has orders of magnitude fewer parameters than AASIST and no explicit temporal or spectral modeling of its own.

3.3 Augmentation

All non-baseline models are trained with RawBoost (Tak et al., 2022) (convolutive plus additive raw-waveform noise, applied stochastically per batch). We separately test cropping each training utterance to a random 1–4 s window before repeat-padding ("short-crop"), intended to target short-utterance failures (§4).

3.4 Training

All models are fine-tuned end-to-end with cross-entropy loss (bona-fide vs. spoof, class-weighted 3:1 toward spoof), Adam, and an exponential learning-rate schedule, using the frontend's native sample rate (16 kHz) and a repeat-padded 4.0 s window at training time. Checkpoints are selected on a held-out development split; because this split is used for early stopping by every model, we treat its absolute EER as *saturated* (uninformative in absolute terms; see §4) and use the shared task's public development-test leaderboard as our primary validation signal instead.

3.5 Ensembling

Our final system averages per-utterance logits (higher = more bona-fide) across N independently trained members, $s = \frac{1}{\sum_i w_i} \sum_{i=1}^{N} w_i \, \ell_i$, with integer weights $w_i \in \{1, 2\}$ set by a small amount of manual up-weighting of the single strongest member; we found rank- and z-score-normalized fusion gave no consistent improvement over this raw logit average in our setting. Members are chosen to differ along at least one axis (frontend size, back-end, or training-data recipe) rather than being re-runs of the same recipe with a different random seed.

Figure 1 summarizes the full pipeline, including the two data sources we tested and excluded.

4 Experiments and Results

4.1 Setup

We report EER on two signals: the shared task's public *development-test* leaderboard (16,023 utterances; used throughout system development) and the official *blind evaluation* set (144,210 utterances, released only in the final evaluation phase, with a hard cap of ten submissions). We first establish every architectural finding on the development-test leaderboard, then validate the final recipe once on the blind set.

4.2 Main progression

Table 1 shows the development-test EER at each major milestone. Two findings stand out. First, augmentation is the single largest individual lever: RawBoost alone roughly halves EER relative to the unaugmented baseline. Second, back-end choice *reverses* once augmentation is applied: with RawBoost, our AASIST-based single model plateaus at 5.20% even after full convergence and after adding external attack data (via ASVspoof 2019), while replacing AASIST with the far simpler Pool head lowers the same frontend's error to 4.42%, and adding external data on top of Pool reaches our best *single*-model result, 3.27%. We attribute the AASIST plateau to poor cross-domain generalization of graph-attention back-ends, consistent with the benchmark findings of Dowerah et al. (2025).

Beyond the single-model ceiling, only ensembling continued to lower error: widening the ensemble from 2 to 13 architecturally diverse members lowered EER from 3.08% to 2.34% in five successive steps, and this held even when an added member was individually *weaker* than the ensemble it joined (e.g., two members scoring 3.37% and 3.42% in isolation each still lowered the ensemble EER on addition). Re-weighting members after the ensemble saturated, by contrast, produced no further gain (a 3% relative *regression* in one trial), indicating diminishing returns once member diversity — not member count or weighting — is exhausted.

System	EER (%)
Baseline (XLS-R + AASIST, no aug.)	14.65
+ RawBoost	6.54
+ 2-model ensemble	5.51
+ external data (ASVspoof)	5.42
+ post-trained SSL frontend	5.20
Pool head (single, replaces AASIST)	4.42
+ external data (ASVspoof + MLAAD)	**3.27**
Final diverse ensemble (13 members)	**2.34**

Table 1: Development-test EER progression. "Final diverse ensemble" is our submitted development-phase system (§3.5).

4.3 Negative results

Per-dialect analysis of our best single model showed Gulf-dialect utterances as the hardest subgroup and false accepts (spoof scored as bona-fide) outnumbering false rejects roughly 2:1, concentrated in utterances under 2 s. Motivated by this, we tested two interventions targeting exactly this failure mode, both of which *increased* error:

- **Short-crop augmentation** (random 1–4 s training crops, targeting the short-utterance failure): single-model EER rose from 3.27% to 3.49%.
- **External multi-TTS Arabic corpus** (a ∼54k-clip subset of ArFake (Elsetohy et al., 2025), spanning four TTS/voice-conversion systems and multiple dialects including Gulf): single-model EER rose to 3.91%, and adding it to our best ensemble raised ensemble EER from 2.34% to 2.54%.
- **Vocoder-copy-synthesis pseudo-spoof** (bona-fide Gulf/Saudi speech from SADA (Alharbi et al., 2024), resynthesized through a neural vocoder (Siuzdak, 2023) and relabeled spoof, to manufacture additional Gulf-dialect attack data): adding this data to the ensemble raised EER from 2.34% to 2.57%, consistent with Wang and Yamagishi (2024)'s finding that large-scale vocoded data can help as self-supervised pre-training but hurt when injected directly into supervised

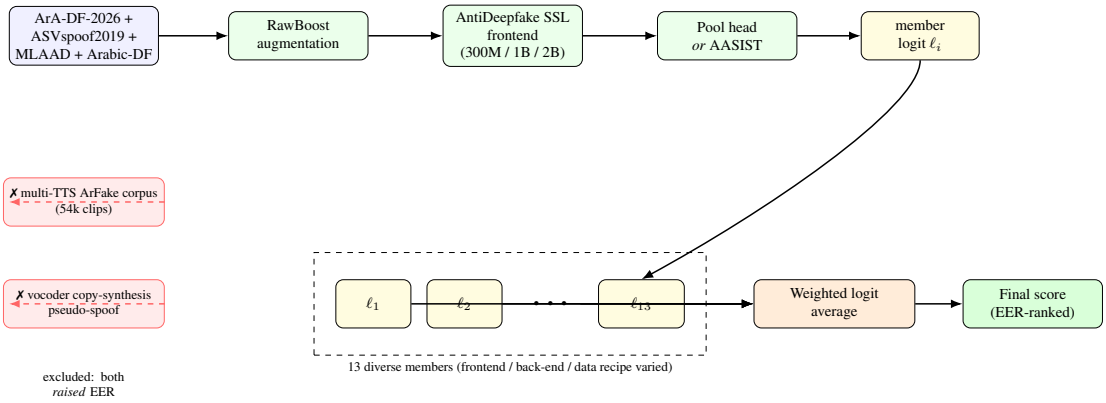


Figure 1: System overview. Each ensemble member is trained with the same augmentation and loss but a different frontend size, back-end, or training-data recipe (top pipeline, repeated 13 times with varied choices). Two additional external data sources (left) were tested and *excluded* after controlled experiments showed they raise EER rather than lowering it (§4.3). Member logits are combined by weighted averaging (bottom right) to produce the submitted score.

fine-tuning (our setting), and with the broader domain-mismatch account of Müller et al. (2022).

We interpret all three as instances of the same failure: augmentation or external data that does not share the target domain's artifact distribution can move a model *away* from the test distribution even when it is topically well-targeted (correct dialect, correct short-duration regime). Notably, a smaller-scale pilot with the same multi-TTS corpus (2.6k clips, i.e. under 5% of that model's training mix) left the ensemble statistically unchanged, suggesting the effect scales with the fraction of mismatched data in the training mix rather than being an all-or-nothing effect.

4.4 Blind evaluation

Our final system (Table 1, last row, re-inferred on the blind split) reached **2.42% EER** on the official 144,210-utterance blind test — close to its 2.34% development-test score, indicating the diverse ensemble generalizes rather than overfitting the smaller development-test set. On the final public blind leaderboard, this ranked our system 8th of the 8 teams listed, with the top-ranked system reported at 0.87% EER — a substantially better score than we were able to reach with the architectures and data available to us, and one we attribute to a materially different training recipe rather than to ensembling or augmentation choices alone.

5 Discussion

Two patterns from our development cycle generalize beyond this shared task. First, a validation split used for checkpoint selection saturates quickly for this kind of high-capacity, SSL-frontend model, so absolute numbers on it stop discriminating between systems; we relied on the public leaderboard as an external, harder validation signal once this saturation was evident. Second, and more surprisingly, more synthetic training data is not a free lunch even when it is topically matched to a known weakness: both of our negative-result interventions specifically targeted Gulf-dialect short utterances and both made that exact subgroup worse, underlining that the choice of *synthesis technique* used to generate extra spoof data matters as much as its topical relevance. These patterns mirror opposite lessons from other Arabic NLP work: general-domain pre-fine-tuning helps for punctuation restoration (Sakr and Torki, 2023), whereas injecting mismatched external data mid-pipeline hurts here, while a continuous dialectness score (Sakr et al., 2024) could sharpen our own coarse (Egyptian/Gulf/MSA) per-dialect error analysis.

6 Conclusion

We presented a Track-1 ArA-DF-2026 system built around three findings: augmentation is the largest single lever before architecture matters, back-end choice (a minimal pooling head over graph attention) determines the single-model ceiling once augmentation is applied, and diverse ensembling is the only lever that keeps paying off after that ceiling

is reached. We also reported two controlled negative results — a large external multi-TTS corpus and a vocoder-based pseudo-spoof corpus both increased error — that we hope are useful cautionary evidence for future participants considering similar data-augmentation strategies.

Limitations

Our study is limited to one shared task's data distribution and label convention; the negative results in §4.3 may not generalize to other spoof corpora or synthesis techniques, and we did not have the compute budget to fully disentangle whether the harm from added data stems from domain mismatch, from diluting the in-domain training signal, or from the specific text-to-speech/vocoder systems used to build that data. Our development-test validation signal comes from a leaderboard with a limited daily submission budget, which constrained the granularity of some ablations (e.g., we could not systematically sweep ensemble weights). Finally, our best single model's absolute performance is well above the best-performing system on the shared task's public leaderboard, so the architectural choices reported here should be read as a solid but not state-of-the-art recipe for this problem.

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